

# Polarization Coil Design

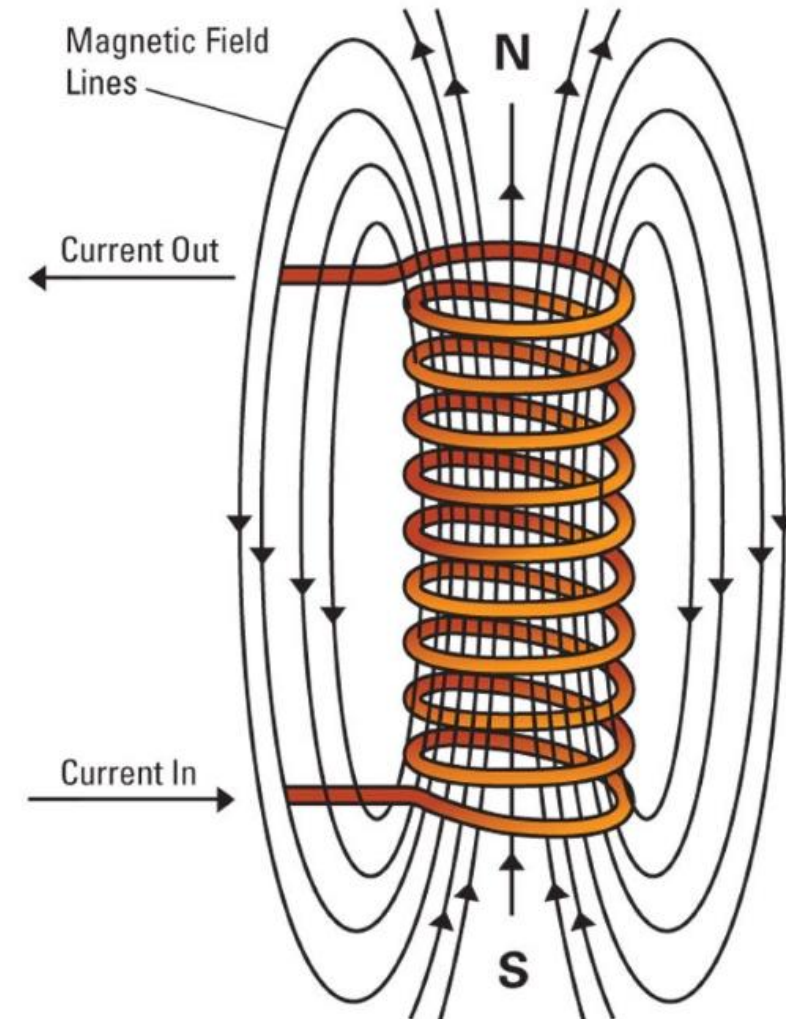
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# Coil Geometry

- My first decision was between these shapes:
  - Single Solenoidal
  - Poly-solenoidal
  - Toroidal

Ideally a toroidal shape would be the best fit, due to its increased independency of alignment with the magnetic fields, and increased resistance to noise. However, due to the difficulty of this geometry, many previous groups bailed their attempts at this.

Typical Solenoidal is more achievable, so we will focus on successfully building this first. Three coil design (one polarization, two sensing) will yield separate design parameters for polarization and sensing coils. Also, sensing coils yield [signal + noise from coil A – noise from coil B], so we will go with that.



# Optimizing the Polarization Coil

- Several sources have mentioned that increasing polarization magnetic field increases the voltage induced by the proton's precession. Also, the greater the liquid content, the greater the voltage induced (more protons contributing).

$$B_{\text{polarize}} \approx \mu_0 \frac{NI}{\ell}, \quad L \approx \mu_0 \frac{N^2 A}{\ell}, \quad R \approx \rho \frac{2\pi r_{\text{mean}} N}{A_{\text{wire}}}$$

- From this, NI can be maximized for maximizing polarization. We should have a current source. Increasing turns does not mean more B field (doubling turns doubles resistance). Increasing turns also increases stored energy, which is not desired.
- Energy stored in polarization coil needs to be released as quickly as possible, so that sensing readings can begin immediately. Increasing length can maximize volume of liquid without increasing energy stored.

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$$V_{\text{FID},0} \approx K B_p (1 - e^{-t_p/T_1})$$

T1 = 0.95 seconds for water

$$U = \frac{1}{2} L I^2$$

$$\mathcal{E} = -N \frac{\Delta \Phi}{\Delta t}$$

- It seems there have been success with amperage on the order of a few amps, so lets start with a 3A for powering the polarization coil. From this, we get:
  - 18 AWG wire is suitable for 3A when only polarized for several seconds
  - And the time constant for growth and decay is  $L/R \approx 3.5\text{ms}$
- Dimensions: Dimensions should be large enough to work with; too large increases  $U = 0.5LI^2$ 
  - The cross section will be 7x14cm. This should be large enough to contain liquid and sensing coils
  - The length should be: 10cm. This should be just large enough to keep the induced flux parallel throughout the fluid
  - Turns: 4 layers of 90 turns (360 total turns). B is proportional to N, but if U too high, reduce N.
  - Wire length: about 144 m
- Characteristics:
  - 3 Ohms
  - 15.4mH
  - 13.6 mT (at 3A)
  - 69mJ energy stored (at 3A)
- Sri Lanka article suggests a polarization time of 5+ seconds
- JPM-4 article mentions decay of precession signal at 0.95s with kerosene. Starts at between 0.5uV (Sri Lanka) to 2uV (JPM-4)

# Discharging the Polarization Coil

- A few DIY projects use TVS-based quenching
- Use the MOSFET to interrupt polarization current and direct it into the clamp path.
- Choose the MOSFET and insulation for the actual clamp voltage, supply voltage, and overshoot.
- Keep the sensing amplifier protected during turnoff, then measure how long ringing takes to settle.

# Optimizing the Sensing Coils

$$\mathcal{E} = -N \frac{\Delta \Phi}{\Delta t}$$

$$V_N = \sqrt{4RkTB}$$

- Since we are going with 3 coil design (two sensing inside a polarization coil), the cross section and height will be easy: 6x6x10cm
- Next is turns and wire gauge
  - Turns increases received voltage, but increases resistance
  - Thinner wire increases noise
  - SNR does not change much between these, so dimensional constraints will decide the design

## Final Design:

- 22 AWG, 138 turns, 4 layers, 6x6cm square, 10cm height, 0.7mm diameter enameled wire, 135m of wire

## Wire Properties per coil:

- 7 ohms, 10-12mH

# Tuning Capacitor for Sensor Coi

$$C = \frac{1}{(2\pi f_0)^2 L}$$

- Finally, resonant tuning is the next part of getting useful voltage from this winding.
- The frequency is Larmor frequency
- $C = 264\text{nF}$  <https://wiki.analog.com/university/courses/alm1k/circuits1/alm-cir-lc-resonator>
- $264\text{nF}$
- Bandwidth will be around 133 Hz
- Polarization ringing time constant will be about 2.4 ms

$$Q = \frac{1}{\omega_0 RC}$$

$$B = \frac{R}{L} = \frac{\omega_0}{Q}$$

$$\tau \approx \frac{Q}{\pi f_0}$$

# Circuit Design

# Considerations

- Since the signal is so small, first stage of amplification should be low noise

# CAD print

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